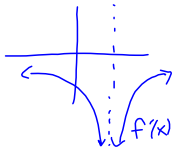
Section 5.1 - The First Derivative

Recall:
- when $f(x)$ is increasing, $f'(x) > 0$
- when $f(x)$ is decreasing, $f'(x) < 0$
- when $f(x)$ is constant, $f'(x) = 0$

Thus, in this section we will be creating sign charts of $f'(x)$ to determine important information about $f(x)$.



Definition: A **partition number** of $f'(x)$ is a value of x such that

$f'(x) = 0$ OR $f'(x)$ is undefined

(these are the only places that $f'(x)$ might possibly change sign)

Definition: A **critical value** of $f(x)$ is a value of x such that

x is a partition number and x is in the domain of $f(x)$

Example 1: Find the critical values and partition numbers for the following functions and then determine where the function is increasing/decreasing.

a) $f(x) = -x^3 + 12x - 5$ Domain: $\mathbb{R}$

$f'(x) = -3x^2 + 12$

$f'(x) = 0$

$-3x^2 + 12 = 0$

$-3(x^2 - 4) = 0$

$-3(x+2)(x-2) = 0$

$x + 2 = 0 \quad x - 2 = 0$

$\boxed{x = -2 \quad x = 2}$ ← partition #'s and critical values

		$x=-3$		-2		$x=0$		2		$x=3$	
$f(x)$	↘				↗				↘		
$f'(x)$		$-$				$+$				$-$	

$\boxed{\text{Decreasing } (-\infty, -2) \cup (2, \infty) \quad \text{Increasing } (-2, 2)}$

b) $g(x) = \dfrac{x+3}{5-x}$ Domain: $x \neq 5$

$g'(x) = \dfrac{(5-x)(1) - (x+3)(-1)}{(5-x)^2} = \dfrac{5 - \cancel{x} + \cancel{x} + 3}{(5-x)^2}$

$= \dfrac{8}{(5-x)^2}$

		$x=0$		5		$x=6$	
$g(x)$	↗				↗		
$g'(x)$		$+$				$+$	

$g'(x)$ is never zero since the numerator is a non zero constant.

$\boxed{\text{Increasing } (-\infty, 5) \cup (5, \infty)}$

$g'(x)$ is undefined when

$(5-x)^2 = 0$

$5 - x = 0$

$\boxed{5 = x}$ ← partition #

$\boxed{\text{No critical values}}$

c) $h(x) = x\ln x - x$ Domain: $x > 0$

$h'(x) = x \cdot \frac{1}{x} + \ln(x) \cdot 1 - 1$
$= \cancel{1} + \ln(x) - \cancel{1}$
$= \ln(x)$

$h'(x) = 0$
$\ln(x) = 0$
$e^0 = x$
$\boxed{1 = x}$ ← partition # and a critical value

$h(x)$	↘		↗
$h'(x)$	−		+
	0	$x = 1/2$	1	$x = 2$

$\boxed{\text{Decreasing } (0,1) \quad \text{Increasing } (1,\infty)}$

Definitions:

- We call $f(c)$ a **relative (local) maximum** if there exists an interval (m,n) containing c such that

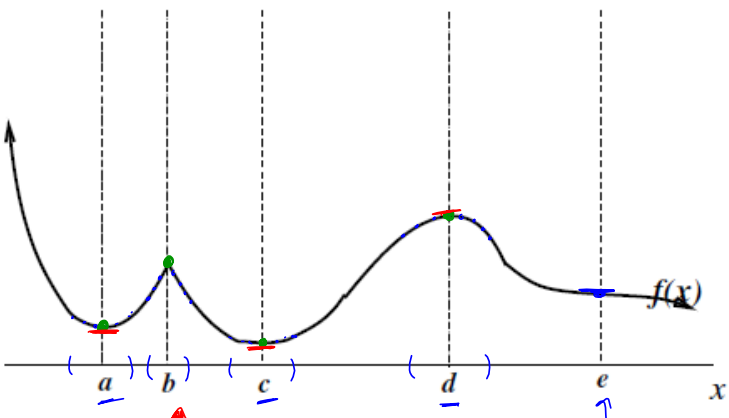
  $f(c) \geq f(x)$ for all x in (m,n)

- We call $f(c)$ a **relative (local) minimum** if there exists an interval (m,n) containing c such that

  $f(c) \leq f(x)$ for all x in (m,n)

- The quantity $f(c)$ is called a **relative (local) extremum** if it is either a local maximum or a local minimum.

Example 2: Let's visualize relative (local) extrema:



Local Min @ $x=a$ & $x=c$

Local Max @ $x=b$ & $x=d$

$f'(a)=0$ $f'(b)$ is undefined $f'(c)=0$ $f'(d)=0$ $f'(e)=0$ but not a local extrema

First Derivative Test: Suppose $x = c$ is a critical value of $f(x)$

1. If $f'(x)$ changes from + to − at $x = c$, then we have that $f(x)$ is ↗↘ and at $x = c$ there is a (relative) local maximum.
2. If $f'(x)$ changes from − to + at $x = c$, then we have that $f(x)$ is ↘↗ and at $x = c$ there is a (relative) local minimum.
3. If the sign of $f'(x)$ is the same on both sides of $x = c$, then at $x = c$ there is neither a local max nor a local min.

Example 3: Determine the intervals where the following functions are increasing and decreasing and find the local extrema.

a) $f(x) = x^4 + 2x^3 + 5$ Domain: $\mathbb{R}$

$f'(x) = 4x^3 + 6x^2$

$f'(x) = 0$

$4x^3 + 6x^2 = 0$

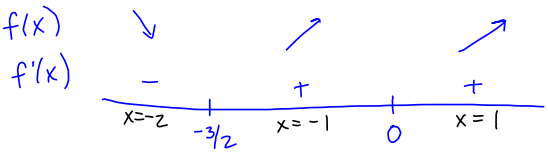
$2x^2(2x+3) = 0$

$2x^2 = 0$ $\quad$ $2x + 3 = 0$

$x = 0$ $\quad$ $2x = -3$

$x = -3/2$

partition #'s & critical values



Decreasing $(-\infty, -3/2)$

Increasing $(-3/2, 0) \cup (0, \infty)$

Local min $(-3/2, 53/16)$

$f(-3/2)$

b) $g(x) = \dfrac{2}{x^2 - 16}$ Domain: $x \neq -4, 4$

$g'(x) = \dfrac{(x^2-16)(0) - 2(2x)}{(x^2-16)^2} = \dfrac{-4x}{(x^2-16)^2}$

$g'(x) = 0$ $\quad$ $g'(x)$ is undefined

$-4x = 0$ $\quad$ $(x^2-16)^2 = 0$

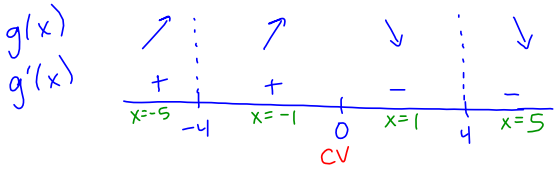
$x = 0$ $\quad$ $x^2 - 16 = 0$

$(x+4)(x-4) = 0$

$x = -4, x = 4$

only critical value

partition #'s



Increasing $(-\infty, -4) \cup (-4, 0)$

Decreasing $(0, 4) \cup (4, \infty)$

Local max $(0, -\frac{1}{8})$

$g(0)$

c) $h(x) = (x+2)e^x$ Domain: $\mathbb{R}$

$h'(x) = (x+2)\cdot e^x + e^x \cdot 1$

$= e^x(x+2+1)$

$= e^x(x+3)$

$h'(x) = 0$

$e^x(x+3) = 0$

$e^x = 0$ no value of x for which $e^x = 0$

$x+3=0$

$x=-3$ partition # and critical value

e^x

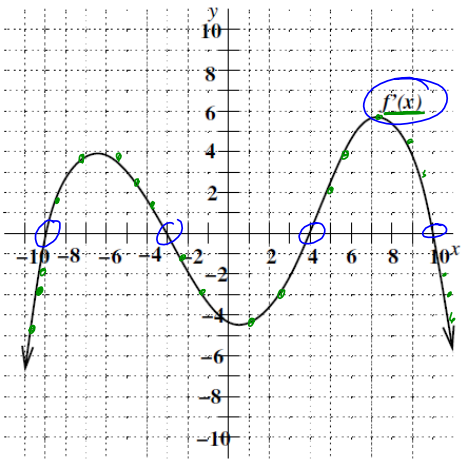
$h(x)$	↘		↗
$h'(x)$	−		+
	$x=-4$	-3 CV	$x=0$

Decreasing $(-\infty, -3)$

Increasing $(-3, \infty)$

Local min $(-3, -e^{-3})$ ← $h(-3)$

Example 4: Given that the domain of $f(x)$ is all real numbers and the graph below is of $f'(x)$, determine the following:



$f(x)$	↘		↗		↘		↗		↘
$f'(x)$	−		+		−		+		−
		-9 CV		-3 CV		4 CV		10 CV	

a) the interval(s) on which $f(x)$ is increasing.

$(-9,-3) \cup (4,10)$

c) Local min @ $x=-9, x=4$

Local max @ $x=-3, x=10$

b) the interval(s) on which $f(x)$ is decreasing.

$(-\infty,-9) \cup (-3,4) \cup (10,\infty)$

c) the critical value(s) (classify each).

Section 5.1 Highly Suggested Homework Problems: 1-41(odd), 45-53(odd), 61, 63, 73, 75, 87, 89